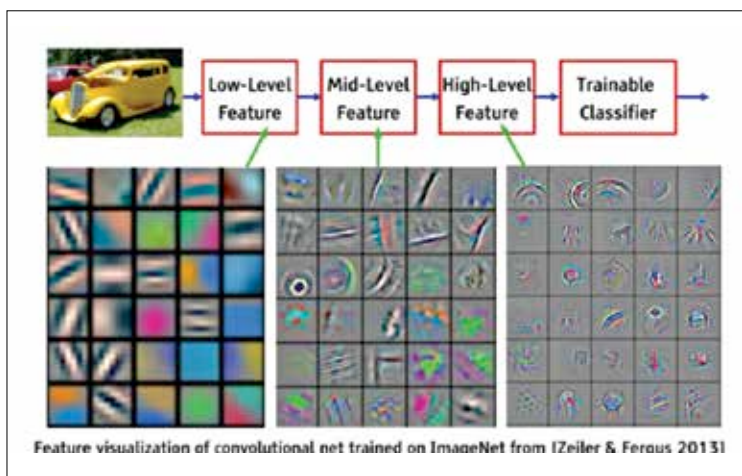


that the deeper you go down the network through various layers, the more image specific features of images are learnt. Let us explain this concept using an example. In a neural network that is trying to detect faces, first the network learns to detect edges in the first layer, then some basic shapes in the second and it learn to detect complex features as it goes deeper and deeper.

All Images have shapes and edges. We can only identify differences between them as soon as we start extracting higher-level features like a nose in a face in the subsequent deeper layers of the network. Only then can we correctly classify an image, because the earlier layers of a neural network will always detect the same basic shapes and edges that are present most of the shapes. Hence, without deep layers, it is unable to classify images correctly most of the time.



Now talking about our problem of differentiating dogs from cats, we can use models that have been trained on huge datasets containing different types of animals. This works because these models have already learnt the basic shape and structure of animals. Therefore, all we are required to do is teach our model the high level features of our new images.

For example, we can use a large neural network like ImageNet that contains about 1.4 million labelled images and 1000 different categories of animals and everyday objects. Since this model already knows how to classify different animals, then we can build upon this existing knowledge to quickly train a new classifier to distinguish between cats and dogs.